

Clinical Decision-Support: Comprehensive Genetic & Dermatological Dossier

PHYSICIAN & MEDICAL GENETICS EDITION | Record ID: MODEL-DERM-VISION | Model: Fitzpatrick Phototype Calibrated (ITA & Melanin Index Decoupling) + Genomic Risk

Patient Demographics	Age: 0 yrs Biological Sex: Architecture Detected Phototype: Multi-Phototype Calibrated	Tri-Model Consensus Risk	89% (MULTI-MODAL DEPLOYED)
Dual Boosting + VQC	XGBoost (0.45) + AdaBoost (0.25) + 4-Qubit VQC (0.30)	Epistemic Uncertainty	Confidence: Validated (5-Fold CV) ($\sigma = \pm 0.108$)

1. Detailed Molecular & Genetic Deficiencies Analysis

Genetic Pathway / Defect	Status / Score	Classification	Clinical Oncologic Implications
TP53 Tumor Suppressor (17p13.1 DNA Binding)	Score: 0.72 / 1.0	PATHOGENIC	Loss of apoptotic checkpoint control, impaired p21/CDKN1A induction, elevated genomic destabilization.
BRCA1 / BRCA2 (HR Repair Pathway)	Presence: 1.0	DEFICIENT (HRD+)	Homologous recombination double-strand break repair failure; high sensitivity to PARP inhibition & platinum agents.
Tumor Mutational Burden (Somatic Density)	12.4 mut/Mb	HIGH (TMB-H)	Elevated somatic neoantigen expression. Biomarker for immune checkpoint inhibitor response (Anti-PD-1/PD-L1).
hs-CRP / Inflammation (Microenvironment)	3.8 mg/L	ELEVATED	Sustained pro-inflammatory cytokines (IL-6 / TNF-alpha) facilitating cutaneous angiogenesis & immune evasion.
Hereditary Pedigree (First-Degree Relatives)	Confirmed (1.0)	POSITIVE	Germline susceptibility inheritance profile. Multi-generational screening indicated.

2. Dual Boosting (XGBoost + AdaBoost) & Parameterized Quantum VQC Ensemble

Inference Engine	Estimated Probability	Weight	Algorithm Architecture & Role
XGBoost (Gradient Boosted Trees)	92%	45%	Depth-6 gradient boosting over combined 17-dimensional cutaneous & genomic vector.
AdaBoost (Adaptive Stumps)	64%	25%	Sequential decision stumps emphasizing borderline cases with ambiguous melanin-genomic interplay.
Quantum VQC (Qiskit Aer / PennyLane)	94%	30%	4-qubit parameterized variational circuit with angle embedding exploring non-linear Hilbert space correlations.
Hybrid Calibrated Consensus	89%	100%	Calibrated consensus with discordance variance boundary (± 0.208). Stratified to HIGH RISK.

3. Model Risk Spectrum Across All Skin Phototypes (Fitzpatrick Types I–VI)

Simulates the patient's exact genetic mutations across all skin pigmentation categories to evaluate cutaneous susceptibility:

Fitzpatrick Phototype	Optical ITA / Melanin	XGBoost	AdaBoost	Hybrid Consensus	Stratification Tier
Type I (Very Light)	ITA: 78° Mel: 3.5	98%	68%	88%	Very High
Type II (Light)	ITA: 52° Mel: 14.0	98%	66%	87%	Very High

Type III (Intermediate)	ITA: 35° Mel: 22.0	98%	64%	86%	Very High
Type IV (Tan / Olive)	ITA: 18° Mel: 30.0	97%	61%	84%	Very High
Type V (Brown)	ITA: -12° Mel: 46.0	94%	57%	82%	Very High
Type VI (Dark / Deeply Pigmented)	ITA: -62° Mel: 72.0	88%	53%	78%	High

4. Actionable Clinical Guidance & Triage Protocol

Primary multimodal screening engine for cutaneous malignancy triage.

MANDATORY MEDICAL DISCLAIMER: AI-generated decision-support output. Not a final medical diagnosis. Final clinical decision remains with a qualified healthcare professional.